

# ServiceNow Metro Ticket Booking System

Project Architecture, Stakeholder Personas & Workflow Design Specification

**Overview:** In ServiceNow terminology, Phase 1 is defined as the **Ideation / Inception Phase**. Brainstorming focuses on defining core business objectives, persona mapping, data architecture, catalog variables, and end-to-end fulfillment flows.

## 1. Brainstorming Deliverables

### A. Core Problem & System Goals

- **Problem Statement:** Users face long physical queues at transit counters and lack a unified enterprise platform to book, track, or seek reimbursement for daily transit tickets.
- **System Objective:** Enable users to dynamic select stations, automatically compute distance-based fares, issue digital tickets with QR codes, and trigger background fulfillment/approvals via ServiceNow Service Portal.

### B. Target User Personas

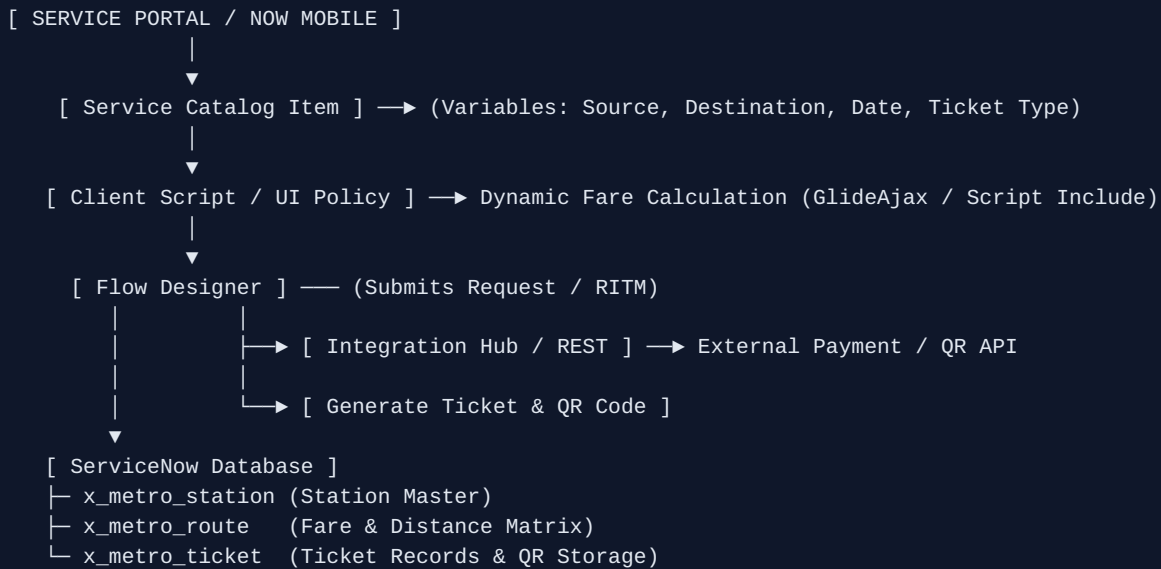
| Persona                | Role & Responsibilities                                                        | Platform Touchpoint                  |
|------------------------|--------------------------------------------------------------------------------|--------------------------------------|
| Commuter (End User)    | Selects route, checks dynamic fare, places order, and views active QR tickets. | Service Portal / Now Mobile App      |
| Metro Operations Admin | Manages station records, fare rules, schedules, and active routes.             | ServiceNow Workspace / Custom Tables |
| Finance / Approver     | Validates corporate commute allowances or payment gateway reconciliation.      | Approval Hub / RITM Flow             |
| Fulfillment Officer    | Validates QR codes at entry/exit gates or overrides ticket status.             | REST API / Mobile Scanner            |

### C. Data Model Architecture (Tables)

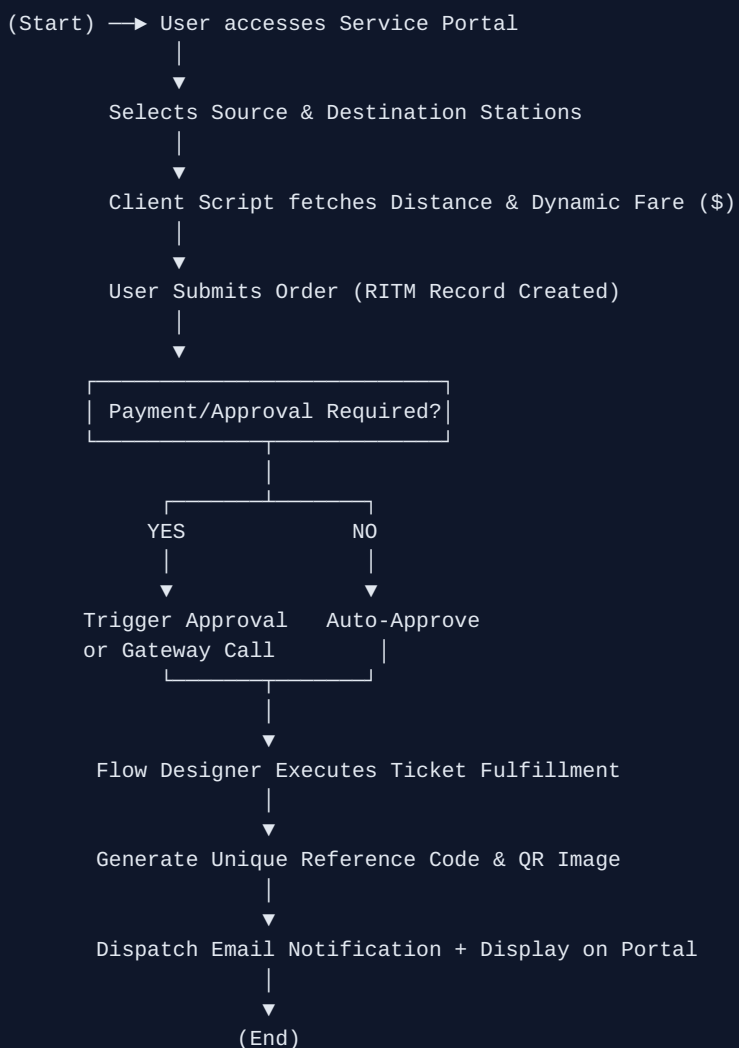
| Table Name          | Technical Name      | Description & Fields                                                                 |
|---------------------|---------------------|--------------------------------------------------------------------------------------|
| Station Master      | x_metro_station     | Stores Station Name, Station Code, Zone, Line Color, Active Status.                  |
| Route & Fare Matrix | x_metro_route       | Maps Source & Destination, Distance (km), Zone tier, and Base Fare (\$).             |
| Catalog Item        | sc_cat_item         | "Book Metro Ticket" item exposed on the Service Portal interface.                    |
| Requested Item      | sc_req_item         | RITM record storing individual ticket transactions and flow states.                  |
| Ticket Master       | x_metro_pass_ticket | Stores Ticket Reference, QR Code Base64/URL, Expiry, Status (Issued, Used, Expired). |

## 2. Visual System Diagrams

### A. System & Conceptual Architecture



### B. End-to-End User & Ticket Journey Workflow



### 3. Brainstorming Implementation Checklist

- ☐ **Catalog Variables:** Configure Source Station, Destination Station, Travel Date, Ticket Type (Single / Day Pass), Quantity.
- ☐ **Catalog Scripts:** Implement onChange scripts + GlideAjax to fetch prices without reloading.
- ☐ **Validation Rules:** Prevent identical Source & Destination selections using UI Policies / Client Scripts.
- ☐ **Flow Designer Logic:** Configure automated approvals for standard passes and route generation for QR triggers.
- ☐ **Integration Hub:** Set up REST step for QR code generation API / PDF ticket generator attachment.